

CCNA Routing Concepts — Important MCQ

◆ Part A: Router & Network Fundamentals

1. What is the primary function of a router?

- A. Connect devices within the same LAN
- B. Determine the best path and forward packets
- C. Assign IP addresses automatically
- D. Convert domain names to IP addresses

Answer: B

2. Which device is responsible for forwarding packets from one network to another?

- A. Hub
- B. Switch
- C. Router
- D. Repeater

Answer: C

3. Which of the following describes a physical topology?

- A. Path over which data is transferred
- B. Arrangement of cables and network devices
- C. Routing table structure
- D. IP addressing scheme

Answer: B

4. What does logical topology describe?

- A. Physical arrangement of cables
- B. Device hardware configuration

- C. Path over which data is transferred
- D. Type of network cable

Answer: C

5. Which characteristic of a network indicates how easily it can accommodate additional users and data?

- A. Reliability
- B. Availability
- C. Scalability
- D. Security

Answer: C

6. Network reliability is often measured using:

- A. RTT
- B. MTBF
- C. TTL
- D. Mbps

Answer: B

7. A Cisco router uses which operating system?

- A. Windows Server
- B. Linux
- C. Cisco IOS
- D. Unix

Answer: C

8. Which of the following is volatile memory?

- A. ROM
- B. Flash

- C. NVRAM
- D. RAM

Answer: D

9. Which memory retains its contents without constant power?

- A. RAM
- B. ROM
- C. Volatile memory
- D. CPU cache

Answer: B

10. Multiple networks connected to a router require:

- A. Multiple CPUs
- B. Multiple interfaces
- C. Multiple operating systems
- D. Multiple routing protocols

Answer: B

Part B: Default Gateway & Host Connectivity

11. What information does a host normally require for network access?

- A. IP address, subnet mask, default gateway
- B. MAC address, DNS name, port number
- C. VLAN ID, SSID, MAC address
- D. Router ID, AS number, metric

Answer: A

12. If a destination device is on the same IP network, what happens?

- A. The packet must go through a router
- B. The packet is sent to the default gateway
- C. The host forwards it directly to the destination
- D. The packet is sent to the ISP

Answer: C

13. If a destination is on a different IP network, a host forwards the packet to:

- A. DNS server
- B. Default gateway
- C. DHCP server
- D. Switch

Answer: B

14. What is the main purpose of a default gateway?

- A. Assign IP addresses
- B. Resolve MAC addresses
- C. Route traffic from a local network to remote networks
- D. Encrypt network traffic

Answer: C

15. Which device normally acts as the default gateway for a host?

- A. Router
- B. Printer
- C. Hub
- D. Access point only

Answer: A

16. Which protocol can dynamically provide a host with an IP address and default gateway?

- A. FTP
- B. DHCP
- C. ARP
- D. ICMP

Answer: B

17. Which devices commonly use static IP addressing according to the chapter?

- A. Smartphones only
- B. Servers and printers
- C. All PCs
- D. Routers only

Answer: B

18. Which information is normally supplied by a DHCP server?

- A. IP address
- B. Subnet mask
- C. Default gateway
- D. All of the above

Answer: D

Part C: Router Configuration

19. Which command is used to assign a name to a Cisco router?

- A. name
- B. hostname
- C. router-name
- D. device-name

Answer: B

20. The `hostname` command is used in:

- A. User EXEC mode
- B. Privileged EXEC mode
- C. Global configuration mode
- D. Interface mode

Answer: C

21. Which command configures a message displayed to users accessing the device?

- A. `message motd`
- B. `banner motd`
- C. `login banner`
- D. `display banner`

Answer: B

22. Which command saves the running configuration into startup configuration?

- A. `save config`
- B. `copy startup-config running-config`
- C. `copy running-config startup-config`
- D. `write running`

Answer: C

23. Why should a router's configuration be saved?

- A. To increase CPU speed
- B. To preserve configuration after reboot
- C. To increase bandwidth
- D. To enable DHCP

Answer: B

24. Which command activates a Cisco router interface?

- A. enable
- B. activate
- C. no shutdown
- D. interface up

Answer: C

25. An IPv4 router interface normally requires:

- A. IP address and subnet mask
- B. MAC address only
- C. Default gateway only
- D. DNS address only

Answer: A

26. In a lab environment, which interface may require a clock rate?

- A. Ethernet interface
- B. DCE serial interface
- C. Loopback interface
- D. Console interface

Answer: B

27. Which command is used to configure an IPv6 address on an interface?

- A. ipv6 address
- B. ip address v6
- C. address ipv6
- D. ipv6 configure

Answer: A

28. Which command can enable an IPv6 link-local address on an interface automatically?

- A. ipv6 link-local
- B. ipv6 enable
- C. ipv6 address auto
- D. enable ipv6

Answer: B

Part D: Loopback Interface

29. What type of interface is a loopback interface?

- A. Physical interface
- B. Wireless interface
- C. Logical interface
- D. Serial interface

Answer: C

30. A loopback interface is commonly used for:

- A. Testing and management
- B. Connecting a physical cable
- C. Providing power
- D. Switching Ethernet frames

Answer: A

31. A loopback interface is:

- A. Connected to a physical port
- B. Internal to the router
- C. Connected directly to a switch
- D. Only available on switches

Answer: B

32. Why can a loopback address be useful for OSPF identification?

- A. It always provides Internet access
- B. It is less likely to go down than a physical interface
- C. It increases bandwidth
- D. It replaces the routing table

Answer: B

Part E: Verification Commands

33. Which command displays a summary of all interfaces including IPv4 addresses and operational status?

- A. show interfaces
- B. show ip route
- C. show ip interface brief
- D. show running-config

Answer: C

34. Which command displays the IPv4 routing table?

- A. show route
- B. show ip route
- C. show routing
- D. show ip routing-table

Answer: B

35. Which command displays detailed information about interfaces and packet flow counts?

- A. show interfaces
- B. show ip route

- C. show history
- D. show ip brief

Answer: A

36. Which command verifies IPv6 routes?

- A. show ipv6 route
- B. show ip route v6
- C. show route ipv6
- D. ipv6 show route

Answer: A

37. In `show ipv6 interface brief`, an interface showing `up/up` indicates:

- A. Only Layer 3 is working
- B. Layers 1 and 2 are operational
- C. Only Layer 2 is working
- D. The routing protocol has converged

Answer: B

38. Which command can be used to test network connectivity?

- A. ping
- B. show
- C. copy
- D. terminal

Answer: A

39. What does the pipe `|` symbol allow when used with Cisco show commands?

- A. Reboot the router
- B. Filter command output
- C. Save configuration
- D. Enter interface mode

Answer: B

40. What does the following command do?

```
show ip interface brief | include down
```

- A. Shows interfaces that are up
- B. Shows only output containing "down"
- C. Removes down interfaces
- D. Shuts down interfaces

Answer: B

Part F: Packet Forwarding — Very Important

41. What happens first when a router receives an Ethernet frame that must be routed?

- A. It changes the IP address
- B. It de-encapsulates the Layer 2 frame
- C. It changes the source IP
- D. It sends an ARP request immediately

Answer: B

42. After de-encapsulation, the router examines:

- A. Destination IP address
- B. Source MAC address only
- C. Destination port
- D. DNS name

Answer: A

43. After finding the best route, the router:

- A. Discards the packet
- B. Encapsulates the Layer 3 packet into a new Layer 2 frame
- C. Changes the destination IP address
- D. Sends the original frame unchanged

Answer: B

44. Which address normally remains unchanged from source to destination?

- A. Layer 2 destination address
- B. Layer 2 source address
- C. Layer 3 IP address
- D. MAC address

Answer: C

45. What happens to Layer 2 addresses at every router hop?

- A. They remain unchanged
- B. They change
- C. They are removed permanently
- D. They become IP addresses

Answer: B

46. Which sequence correctly describes router forwarding?

- A. Encapsulate → route → de-encapsulate
- B. De-encapsulate → examine destination IP → re-encapsulate
- C. Route → encrypt → encapsulate
- D. ARP → change IP → forward

Answer: B

47. PC1 wants to send a packet to a host on a different network. What does PC1 need first to send the Ethernet frame?

- A. Destination server's MAC address
- B. Default gateway's MAC address
- C. ISP's MAC address
- D. DNS server's MAC address

Answer: B

48. Which protocol can PC1 use to discover the MAC address of its default gateway?

- A. DHCP
- B. DNS
- C. ARP
- D. FTP

Answer: C

49. If the default gateway's MAC address is not present in the ARP cache, the host sends:

- A. DNS query
- B. ARP request
- C. DHCP request
- D. ICMP redirect

Answer: B

50. Ethernet Type 0x800 indicates that the Ethernet frame contains:

- A. IPv6 packet
- B. ARP packet
- C. IPv4 packet
- D. TCP segment

Answer: C



Part G: Routing Decisions & Best Path

51. A routing table search can result in how many basic path determinations?

- A. 2
- B. 3
- C. 4
- D. 5

Answer: B

52. If the destination network is directly connected to the router, the packet is:

- A. Sent to another router
- B. Sent to Gateway of Last Resort
- C. Forwarded out the directly connected interface
- D. Dropped

Answer: C

53. If the destination belongs to a remote network, the router:

- A. Sends the packet to another router
- B. Sends it to the local switch
- C. Drops it immediately
- D. Sends it to DNS

Answer: A

54. If no route is found for the destination, the packet may be sent to:

- A. DHCP server
- B. Gateway of Last Resort
- C. DNS server
- D. Loopback interface

Answer: B

55. The best path is generally the path with:

- A. Highest metric
- B. Lowest metric
- C. Highest bandwidth only
- D. Lowest IP address

Answer: B

56. What is a metric?

- A. Router's MAC address
- B. A value used to measure distance to a network
- C. IP address of next-hop router
- D. Administrative password

Answer: B

57. Which metric does RIP use?

- A. Cost
- B. Bandwidth
- C. Hop count
- D. Delay

Answer: C

58. OSPF primarily uses:

- A. Hop count
- B. Cost based on cumulative bandwidth
- C. Reliability only
- D. Number of interfaces

Answer: B

59. Which factors are used by EIGRP according to the chapter?

- A. Bandwidth, delay, load, reliability
- B. Hop count only
- C. Cost only
- D. MAC address and bandwidth

Answer: A

60. Which statement about routing metrics is correct?

- A. Higher metric is always preferred
- B. Lower metric is generally preferred
- C. Metric has no role in routing
- D. Metric is the same for every routing protocol

Answer: B

Part H: Load Balancing & Administrative Distance

61. What happens when a router has two paths with identical metrics to the same destination?

- A. It always selects the first path
- B. It can forward traffic using both paths
- C. It drops the packet
- D. It uses only the path with the highest bandwidth

Answer: B

62. Equal-cost paths used simultaneously are called:

- A. Unequal routing
- B. Equal cost load balancing

- C. Default routing
- D. Static balancing

Answer: B

63. Which routing protocol supports unequal-cost load balancing?

- A. RIP
- B. OSPF
- C. EIGRP
- D. ARP

Answer: C

64. What does Administrative Distance (AD) represent?

- A. Physical distance
- B. Trustworthiness of a route source
- C. Number of routers in a path
- D. Bandwidth of a link

Answer: B

65. Which AD value is preferred?

- A. Higher AD
- B. Lower AD
- C. AD does not matter
- D. Zero only

Answer: B

66. If multiple routing protocols provide routes to the same destination, Cisco IOS uses _____ to select the route source.

- A. Metric
- B. Bandwidth

- C. Administrative Distance
- D. Hop count

Answer: C

Part I: Routing Table

67. Where is the routing table stored?

- A. ROM
- B. Flash
- C. RAM
- D. NVRAM

Answer: C

68. A routing table contains information about:

- A. Directly connected routes only
- B. Remote routes only
- C. Directly connected and remote routes
- D. MAC addresses only

Answer: C

69. Which command displays the IPv4 routing table?

- A. `show ip route`
- B. `show route ip`
- C. `display ip route`
- D. `show routing-table`

Answer: A

70. Which code represents a directly connected route?

- A. S
- B. D
- C. C
- D. O

Answer: C

71. Which code represents a local route?

- A. L
- B. C
- C. S
- D. O

Answer: A

72. Which code represents a static route?

- A. D
- B. O
- C. C
- D. S

Answer: D

73. Which code represents an EIGRP-learned route?

- A. E
- B. D
- C. O
- D. S

Answer: B

74. Which code represents an OSPF-learned route?

- A. O
- B. D

- C. S
- D. C

Answer: A

75. Which of the following is NOT normally a routing table entry detail?

- A. Route source
- B. Destination network
- C. Next-hop address
- D. User's password

Answer: D

76. What does the next-hop address tell the router?

- A. The final destination's MAC address
- B. The next router to forward the packet to
- C. The source host's IP address
- D. The DNS server address

Answer: B

77. What does the outgoing interface specify?

- A. Where the packet should exit the router
- B. Where the packet originated
- C. The source MAC address
- D. The routing protocol

Answer: A



Part J: Directly Connected Routes

78. Before an interface is added as a directly connected route, it must have:

- A. Only an IP address
- B. IP address, be activated, and receive carrier signal
- C. Only a MAC address
- D. Only a default gateway

Answer: B

79. Which command activates an interface?

- A. `enable interface`
- B. `interface up`
- C. `no shutdown`
- D. `router enable`

Answer: C

80. With IOS 15 and later, an active directly connected interface creates:

- A. One routing table entry
- B. Two routing table entries
- C. Three routing table entries
- D. No routing table entry

Answer: B

81. The two entries created by an active directly connected interface are:

- A. S and O
- B. C and L
- C. D and S
- D. O and C

Answer: B

82. In an IPv6 routing table, an `L` entry represents:

- A. A local route with /128 prefix
- B. A link-state route

- C. A long-distance route
- D. A loopback-only route

Answer: A

Part K: Static Routing

83. Static routes are:

- A. Automatically learned
- B. Manually configured
- C. Learned through ARP
- D. Created by DHCP

Answer: B

84. What is one major disadvantage of static routes?

- A. They consume excessive bandwidth
- B. They cannot be manually configured
- C. They must be manually reconfigured when topology changes
- D. They require DHCP

Answer: C

85. Which is a benefit of static routing?

- A. Automatic topology adaptation
- B. Improved security
- C. Automatic route discovery
- D. Automatic load balancing

Answer: B

86. Static routing generally uses:

- A. More bandwidth to exchange routes
- B. No bandwidth for dynamic route updates
- C. DHCP broadcasts
- D. ARP broadcasts continuously

Answer: B

87. Which command configures an IPv4 static route?

- A. `route ip network mask`
- B. `ip route network mask {next-hop-ip | exit-intf}`
- C. `static ip route`
- D. `router static`

Answer: B

88. What code represents a static route in the routing table?

- A. C
- B. L
- C. S
- D. D

Answer: C

89. Which command represents an IPv4 default static route?

- A. `ip route 255.255.255.255 0.0.0.0`
- B. `ip route 0.0.0.0 0.0.0.0`
- C. `ip default-route 0.0.0.0`
- D. `default ip route`

Answer: B

90. A default static route is used when:

- A. The destination is directly connected
- B. No specific route exists in the routing table
- C. ARP fails
- D. DHCP fails

Answer: B

Part L: IPv6 Static & Dynamic Routing

91. What is the IPv6 default static route?

- A. `ipv6 route 0.0.0.0/0`
- B. `ipv6 route ::/0`
- C. `ipv6 default 0/0`
- D. `ipv6 route 255.255.255.255`

Answer: B

92. Which command verifies IPv6 static routes?

- A. `show ip route`
- B. `show ipv6 route`
- C. `show route ipv6 static`
- D. `ipv6 show route`

Answer: B

93. Unlike IPv4, an IPv6 default route does NOT explicitly use:

- A. `::/0`
- B. An interface
- C. Next-hop address
- D. An asterisk/Gateway of Last Resort notation

Answer: D

94. Dynamic routing protocols are primarily used to:

- A. Assign MAC addresses
- B. Share reachability and status information about remote networks
- C. Configure passwords
- D. Replace Ethernet

Answer: B

95. What is a major advantage of dynamic routing over static routing?

- A. It never uses CPU
- B. It can determine a new best path after a topology change
- C. It requires manual configuration of every route
- D. It cannot detect failures

Answer: B

96. When routers have finished exchanging and updating routing information, they are said to have:

- A. Encapsulated
- B. Converged
- C. Switched
- D. Fragmented

Answer: B

97. Which of the following is NOT listed as an IPv4 routing protocol in the chapter?

- A. EIGRP
- B. OSPF
- C. RIP
- D. HTTP

Answer: D

98. Which protocol uses hop count as its metric?

- A. OSPF
- B. EIGRP
- C. RIP
- D. ARP

Answer: C

99. Which routing protocol uses cost based on cumulative bandwidth according to the chapter?

- A. RIP
- B. OSPF
- C. EIGRP
- D. DHCP

Answer: B

100. Which statement best describes dynamic routing?

- A. Routes are manually entered and never change
- B. Routers automatically learn remote networks from other routers
- C. Only directly connected networks are used
- D. Routing tables are stored only in ROM

Answer: B